

M Formula Language

INTERMEDIATE POWER QUERY IN EXCEL



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Advanced Editor

- Written in M formula language (M code)
 - Case-sensitive
- **Let** expressions have two parts:
 - "let" and "in" clause
 - Comma-separated list of variable names or step identifiers.
 - Each step builds upon previous step by referring to its name
 - Step listed after the "in" clause is the visible query output

```
let  
    Source = ""  
in  
    Source
```

Advanced Editor

```
let
    MyTable = Table.FromRecords({
        [ID = 1, Name = "A", Result = 10],
        [ID = 2, Name = "B", Result = 5]})
in
    MyTable
```

Let expression with 1 step

Advanced Editor

```
let
    MyTable = Table.FromRecords({
        [ID = 1, Name = "A", Result = 10],
        [ID = 2, Name = "B", Result = 5]}),
    #"Sorted Rows" = Table.Sort(MyTable,{{"Result", Order.Ascending}})
in
    #"Sorted Rows"
```

Let expression with two steps

Advanced Editor

```
let
    MyTable = Table.FromRecords({
        [ID = 1, Name = "A", Result = 10],
        [ID = 2, Name = "B", Result = 5]}),
    #"Sort" = Table.Sort(MyTable,{{"Result", Order.Ascending}}),
    #"Lower" = Table.TransformColumns(#"Sort",{{"Name", Text.Lower, type text}})
in
    #"Lower"
```

Let expression with three steps

Value types

Primitive

- Single-part value (e.g. number, logical, text, or null)

```
[ID = 1, Name = "A"]
```

Structured

- List
- Record
- Table
- Function

Structured value types

List:

- Zero-based ordered sequence of values with one column

```
List = {1,2,3}
```

Record:

- A structure with a single row, but multiple columns

```
Record = [Column1=1,Column2=2]
```

Table:

- Combination of multiple rows and multiple columns

```
Table = #table({"Column A","Column B"},  
              {{1,10},{2,20}})
```


M Built-in Functions

Function:

- Value that, when invoked with arguments, produces a new value.
- Table functions
- List functions
- Number functions
- Record functions
- Date functions
- Time functions
- Duration functions
- ... many more!

¹ <https://learn.microsoft.com/en-us/powerquery-m/power-query-m-function-reference>

Custom Functions

- Complex calculations
- Reusability
- Parameterization
- Performance optimization

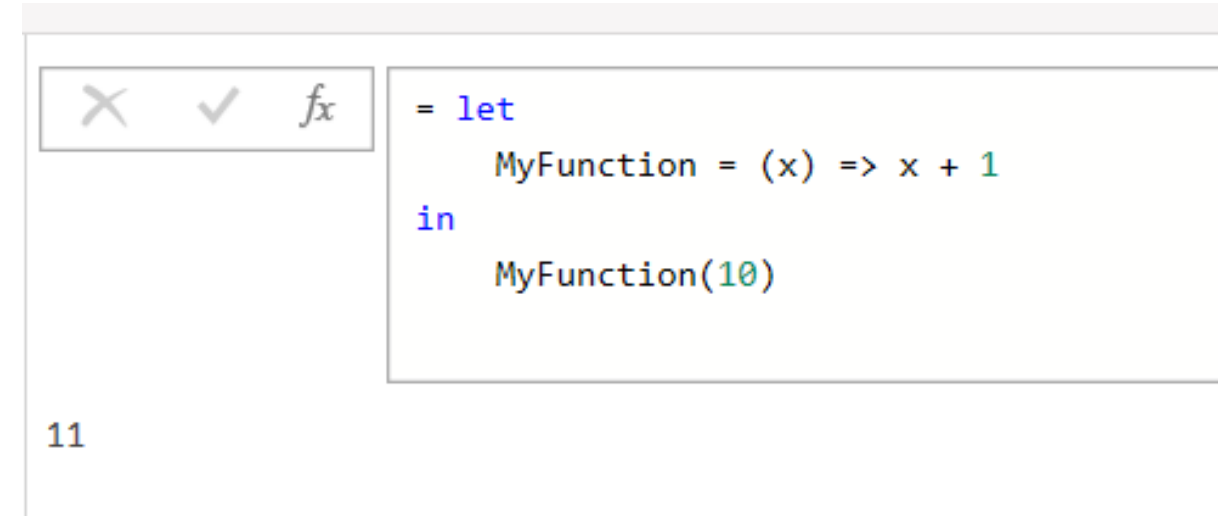
Custom function syntax:

```
= (Variable as Data Type, Variable as Data Type) => (Output Expression)
```

Custom Functions

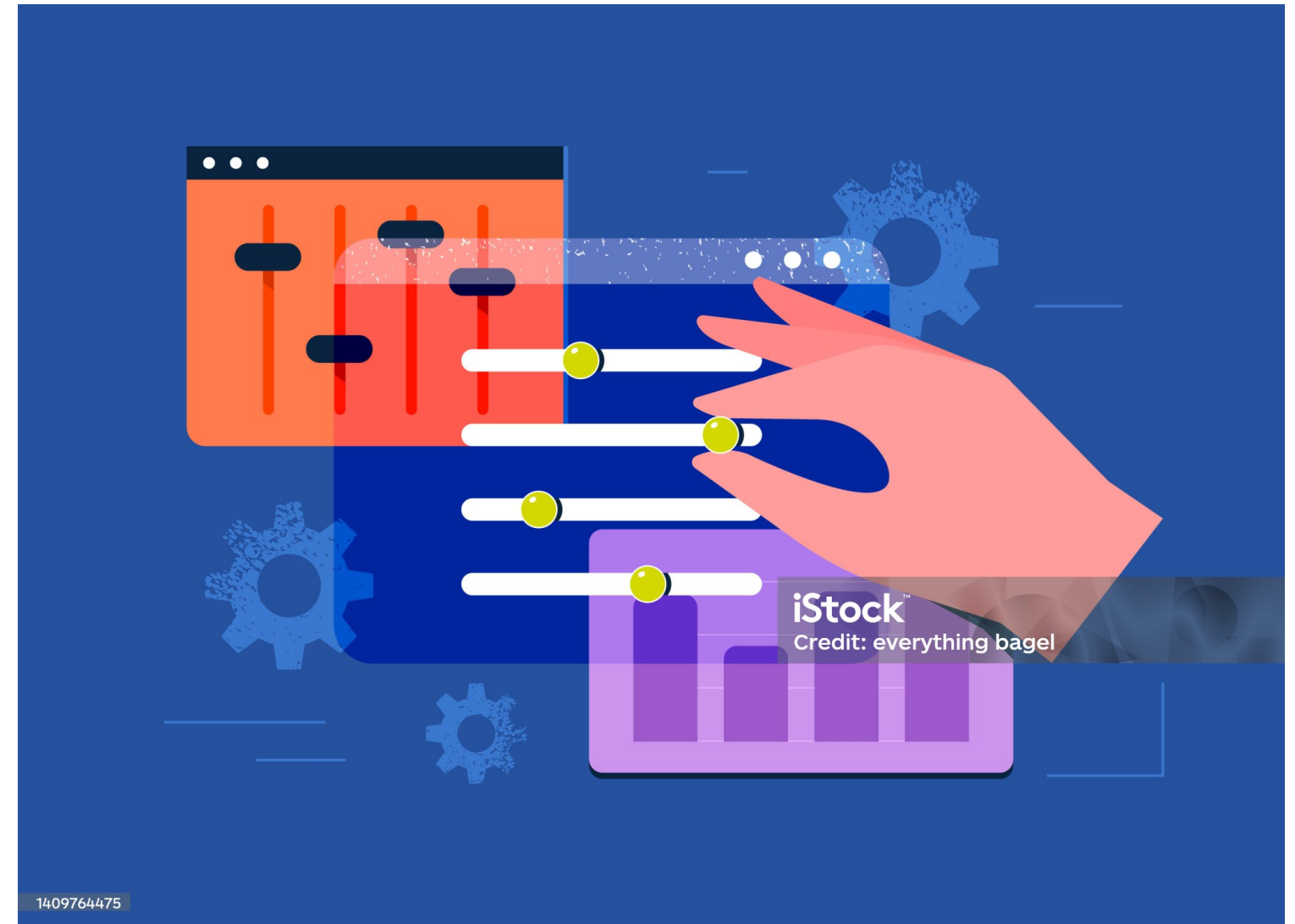
Simple custom function example:

```
let  
    MyFunction = (x) => x + 1  
in  
    MyFunction(10)
```



Query Parameters

- Placeholders to dynamically pass values to a query
 - Make queries more flexible and reusable
 - Dynamic filters
 - Custom functions



Let's practice!

INTERMEDIATE POWER QUERY IN EXCEL

M for Power Query in Excel

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INTERMEDIATE POWER QUERY IN EXCEL

Congratulations!

INTERMEDIATE POWER QUERY IN EXCEL

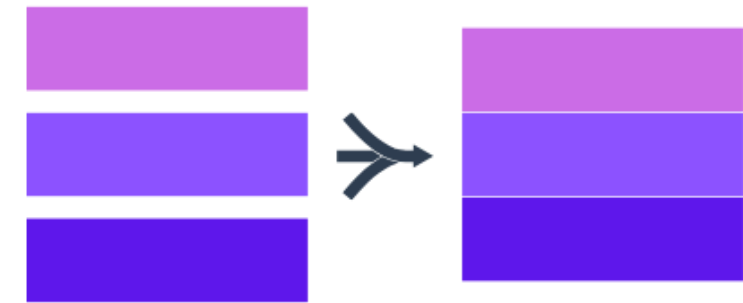


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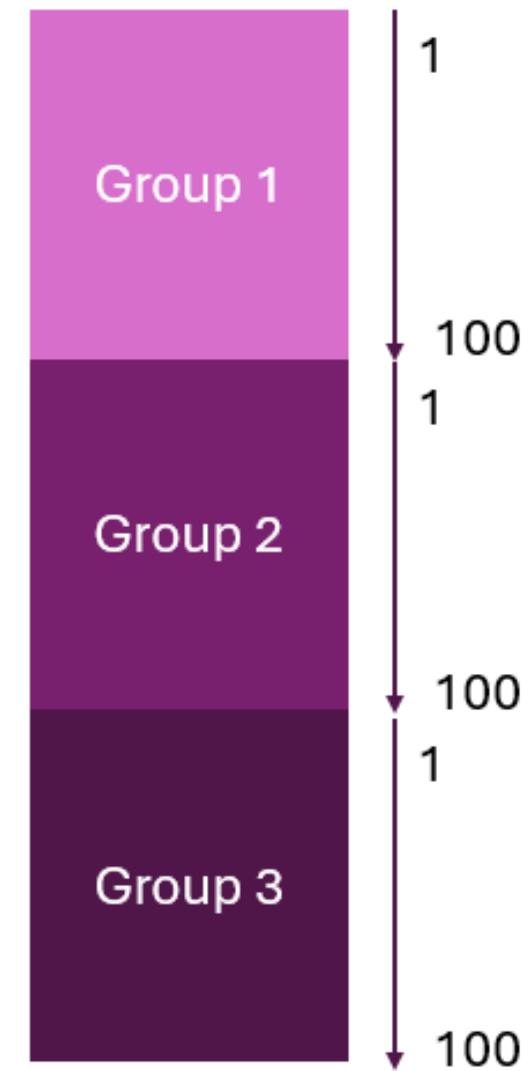
Combining queries in Power Query

- Appending queries
- Merging queries
- Join types
- Special join conditions



Custom Columns

- How M code is generated
- Custom Columns
- Nested Conditional logic
- Advanced indexing



M Formula Language

```
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in
    MyTable
```

- Advanced Editor
- Value types
- Built-in functions
- Query Parameters
- Custom Functions

Congratulations!

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